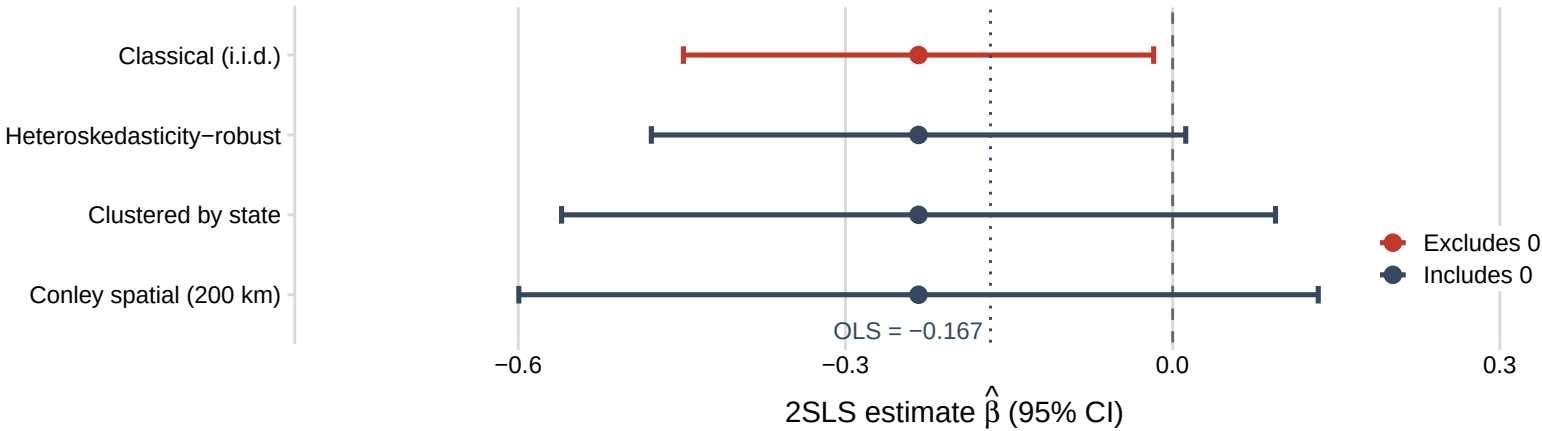


The Geography of Incomplete Recovery

Recession severity, the COVID–19 boom, and the limits of market self–correction

Did the hardest–hit U.S. housing markets catch up during the 2020–2021 boom, or does the apparent catch–up dissolve under credible identification?

1. The headline: an apparent catch–up that vanishes under credible identification

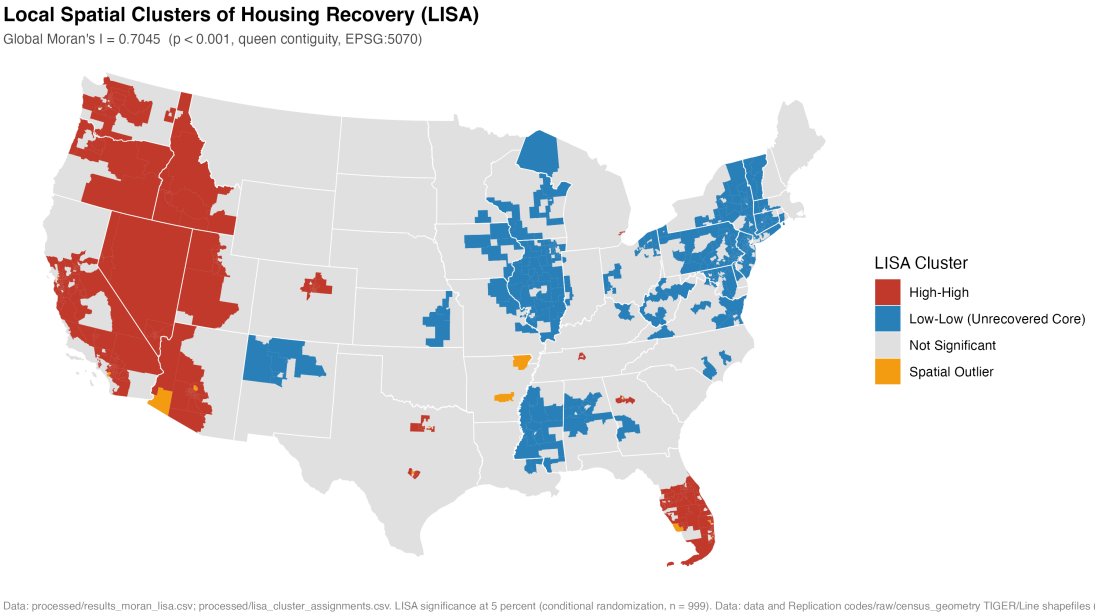


County 2SLS (2,709 counties), two instruments, first–stage $F = 22.7$. $\beta = -0.233$ keeps the OLS sign; it is not a reversal. Significant only under naive errors; once the strong spatial dependence (Moran's $I = 0.70$) is honored it is indistinguishable from zero, and a weak–instrument–robust Anderson–Rubin test cannot reject $\beta = 0$ ($p = 0.07$).

2. Why the apparent catch–up is a mirage that cannot be pinned down

The OLS catch–up ($\beta = -0.167$) is what mean reversion and measurement error produce even with no causal effect. Testing it needs an instrument, and no available one is both strong and valid. The Bartik industry–mix shock is strong ($F = 21$) but INVALID: it fails a pre–trend placebo (it predicts the 2012–2019 housing trend it should not), because Rotemberg weights show it is ~87% construction employment, which is the housing cycle itself. The HMDA high–cost lending share ($F = 28$) passes the placebo but is too weak: its estimate loses significance under spatial inference. Between a strong–but–invalid and a valid–but–weak instrument, the causal effect is NOT IDENTIFIED.

3. Recovery failure is regional and persistent



Local clusters of recovery (LISA): 360 Low–Low cores in the Great Lakes corridor, Appalachia, and the Mississippi Delta.

Because recovery does not self–correct, this spatial inequality is durable, not transitional.
Policy use: the Low–Low footprint is a ready, reproducible CRA assessment–area targeting screen.

Recovery index 2012–2021 (2,351 PUMAs); causal design 2,709 counties.
Sources: IPUMS ACS, FHFA House Price Index, Census County Business Patterns, HMDA, FEMA National Risk Index, Census TIGER/Line.